6TH GRADE MATH · UNIT 1: RATIOS, RATE, PERCENT AND PROPORTIONS

Bake Sale Boss

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students run a class bake sale. They scale recipes with ratio tables, compare unit prices on store price cards, and use percents to set prices, apply discounts, and track profit toward a $300 goal.

The project connects equivalent ratios, unit rates, and percents to one realistic decision: how to plan a profitable booth.

Standards Addressed

- 6.RP.A.1 Understand the concept of a ratio
- 6.RP.A.2 Understand the concept of a unit rate
- 6.RP.A.3 Use ratio and rate reasoning (tables, unit pricing, percents)
- Honors: percent markup, discount, and percent change

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: show a recipe for 24 cookies; ask how to make 72. Part 1 ratio tables and scaling.
Day 2	Part 2 unit prices with the Store Price Cards; whole-class discussion of best buys.
Day 3	Part 3 pricing and percents; groups build their business plan and booth sign.

Materials

- Student handout with the Recipe Card and Store Price Cards
- Calculators
- Chart paper or poster board for the booth sign
- Optional: real store flyers

How the Three Versions Differ

Level	What changes
CP2	Ratio tables with starters, unit prices for single items, and finding a percent of a price.
CP1	Scales to non-whole batches, compares unit prices across all items, plans total cost, and finds profit as a percent of the goal.

Level	What changes
Honors	Uses unit conversions, optimizes a shopping list under a budget, and works with markup, discount, and percent change.

Watch For (Common Misconceptions)

- Adding the same amount to both parts of a ratio instead of multiplying.
- Dividing the wrong way for a unit price (units per dollar instead of dollars per unit).
- Thinking the lower unit price is always the cheapest total.

Extension Ideas

- Survey classmates to predict how many of each treat will sell.
- Graph the ratio table for flour and cookies as points on a coordinate plane.
- Track real sales and compare them with predictions.

Answer Key – CP2

Part 1: Scale the Recipe

#	Task	Answer
1	Write the ratio of cups of flour to cups of sugar three ways.	2 to 1, 2:1, $\frac{2}{1}$
2	Complete a ratio table to find the flour needed for 48, 72, and 96 cookies.	48 → 4 cups, 72 → 6 cups, 96 → 8 cups
3	How many eggs do you need for 72 cookies?	72 is 3 batches, so 3 × 3 = 9 eggs.
4	You have 10 cups of flour. How many cookies can you make? Show your ratio table.	10 ÷ 2 = 5 batches, 5 × 24 = 120 cookies.

Part 2: Best Buys

#	Task	Answer
5	Find the price per pound for each bag of flour. Which is the better buy?	5 lb: $3.50 ÷ 5 = $0.70 per lb. 10 lb: $6.20 ÷ 10 = $0.62 per lb. The 10 lb bag is the better buy.
6	Find the price per egg for each carton. Which is the better buy?	12 eggs: $2.40 ÷ 12 = $0.20 per egg. 18 eggs: $3.42 ÷ 18 = $0.19 per egg. The 18-egg carton is the better buy.
7	Find the price per box for each pack of treat boxes. Which is the better buy?	25 boxes: $5.00 ÷ 25 = $0.20 per box. 60 boxes: $10.80 ÷ 60 = $0.18 per box. The 60-box pack is the better buy.
8	Tell a friend why the bigger package is not ALWAYS the better buy.	Answers vary: a bigger package can cost more per unit, or you may not use it all before it spoils.

Part 3: Price Your Treats

#	Task	Answer
9	You sell cookies for $0.50 each. How much money do you make if you sell 120 cookies?	120 × $0.50 = $60
10	At the end of the day, cookies are 20% off. What is 20% of $0.50? What is the new price?	20% of $0.50 = $0.10. New price = $0.40.
11	Choose two treats and a price for each. If you sell 100 of each, how much money do you make? Is that more or less than 50% of the $300 goal?	Answers vary. Example: cookies $0.50 and muffins $1.00 → $50 + $100 = $150, which is exactly 50% of $300.

Reflection

#	Prompt	Look for
R1	How did a ratio table help you scale the recipe?	Students describe multiplying both quantities by the same number to keep the ratio equal.

#	Prompt	Look for
R2	Why do you divide to find a unit price?	Dividing the price by the number of units gives the cost for exactly one unit, so prices of different sizes can be compared.

Answer Key – CP1

Part 1: Recipes & Ratios

#	Task	Answer
1	Write the ratio of flour to sugar to eggs. What is the ratio of eggs to cookies in simplest form?	Flour : sugar : eggs = 2 : 1 : 3. Eggs : cookies = 3 : 24 = 1 : 8.
2	How much flour, sugar, and how many eggs are needed for 180 cookies?	180 ÷ 24 = 7.5 batches → 15 cups flour, 7.5 cups sugar, 22.5 → 23 eggs (round up).
3	A muffin recipe uses 3 cups of flour for 12 muffins. Which uses more flour per treat, a cookie or a muffin? How many times as much?	Cookie: $\frac{2}{24} = \frac{1}{12}$ cup. Muffin: $\frac{3}{12} = \frac{1}{4}$ cup. A muffin uses 3 times as much flour.
4	You have 15 cups of flour to share between cookies and muffins. Find one combination that uses all the flour, and give the number of each treat.	Answers vary. Example: 3 batches of cookies (6 cups, 72 cookies) + 3 batches of muffins (9 cups, 36 muffins) = 15 cups.

Part 2: Unit Prices & Shopping

#	Task	Answer
5	Find the unit price of each item on both price cards (per pound, per ounce, per egg, or per box). Circle the better buy for each item.	Flour $0.70 vs $0.62 per lb (10 lb). Sugar $0.80 vs $0.75 per lb (10 lb). Chips $0.24 vs $0.21 per oz (24 oz). Eggs $0.20 vs $0.19 (18). Butter $4.50 vs $4.00 per lb (4 lb). Boxes $0.20 vs $0.18 (60).
6	You need 9 cups of sugar. 1 pound of sugar is about 2.25 cups. How many pounds do you need, and which bag (or bags) should you buy?	9 ÷ 2.25 = 4 lb. Buy one 4 lb bag for $3.20 (the 10 lb bag costs more in total even though its unit price is lower).
7	You need 30 eggs. Find the cheapest way to buy them and the cost.	Two 18-egg cartons (36 eggs) = $6.84; one 18 + one 12 (30 eggs) = $5.82; three 12s (36) = $7.20. Cheapest: 18 + 12 for $5.82.
8	Explain why the better unit price is not always the cheapest choice for your shopping list.	You pay for the whole package; if you need only part of a large package, a smaller package can cost less in total.

Part 3: Pricing & Profit

#	Task	Answer
9	You sell 150 cookies at $0.50 and 40 muffins at $1.25. Find your total sales and your profit.	Sales: 150 × 0.50 + 40 × 1.25 = 75 + 50 = $125. Profit: 125 − 45 = $80.
10	What percent of the $300 goal is your profit? How much more do you need?	$\frac{80}{300}$ ≈ 26.7%. Need $220 more.

#	Task	Answer
11	Design a price list with at least three treats. Predict how many of each you will sell and show that your plan earns at least $\frac{1}{3}$ of the goal at one sale.	Answers vary; check that total sales minus supply cost ≥ $100.

Reflection

#	Prompt	Look for
R1	Explain how equivalent ratios and unit rates are connected.	A unit rate is the equivalent ratio with 1 in the second quantity; both keep the same multiplicative relationship.
R2	How did percents help you decide whether your plan was on track?	Writing profit as a percent of the goal shows progress on a common scale even when amounts change.

Answer Key – Honors

Part 1: Recipe Engineering

#	Task	Answer
1	A customer wants a batch of 30 cookies. Find the amount of each ingredient. Give eggs to the nearest whole egg and explain your rounding choice.	30 ÷ 24 = 1.25 → 2.5 cups flour, 1.25 cups sugar, 3.75 eggs → 4 eggs (round up so the dough is not too dry; 3 eggs would be short).
2	A double-chocolate version keeps the flour : sugar ratio but uses 1.5 cups of sugar per batch. How much flour is needed per batch?	Flour : sugar = 2 : 1, so 1.5 cups sugar needs 3 cups flour.
3	1 cup of flour weighs about 125 grams. How many kilograms of flour do you need for 240 cookies?	240 ÷ 24 = 10 batches → 20 cups → 2,500 g = 2.5 kg.
4	Muffins use 3 cups of flour per batch of 12, and cookies use 2 cups per batch of 24. You have 12 cups of flour and want twice as many cookies as muffins. Using whole batches, how many of each can you make?	Twice as many cookies: 24c = 2(12m), so c = m batches. Flour: 3m + 2m = 5m ≤ 12, so m = 2. Make 2 batches of each: 24 muffins and 48 cookies, using 10 cups with 2 cups left.

Part 2: Shopping Optimization

#	Task	Answer
5	Find the percent savings per unit when you buy the larger chocolate-chip bag instead of the smaller one.	Small: \$0.24 per oz. Large: \$0.21 per oz. Savings $\frac{0.03}{0.24}$ = 12.5%.
6	You need 36 eggs, 8 lb of flour, 4 lb of sugar, and 2 lb of butter. Find the cheapest combination of packages and the total cost.	Eggs: two 18s = \$6.84. Flour: one 10 lb = \$6.20 (cheaper than two 5 lb = \$7.00). Sugar: one 4 lb = \$3.20. Butter: two 1 lb = \$9.00 (cheaper than 4 lb at \$16). Total \$25.24.
7	With the rest of your \$40 budget, how many full 60-packs of treat boxes can you buy? How much money is left?	\$40 − \$25.24 = \$14.76. One 60-pack (\$10.80) → \$3.96 left.
8	Is the larger package ever worse even though its unit price is lower? Use your numbers to explain.	Yes: the 4 lb butter has a lower unit price (\$4.00) but costs \$16 total for butter you do not need, while two 1 lb packages cost \$9.

Part 3: Pricing Strategy

#	Task	Answer
9	You mark up each cookie 150% over its ingredient cost of \$0.20. What is the selling price? Then you give a 20% end-of-day discount. What is the final price, and what is the percent change from the original selling price?	Markup: 0.20 + 1.5(0.20) = \$0.50. Discount: 0.50 × 0.8 = \$0.40. Percent change: −20%.

#	Task	Answer
10	Last sale you sold 160 cookies at $0.50. If you raise the price to $0.60, you predict sales drop by 15%. Compare the revenue for both prices.	$0.50 × 160 = $80. 160 × 0.85 = 136 cookies × $0.60 = $81.60. The higher price earns $1.60 more.
11	Design a pricing plan for three treats (with at least one percent discount) that reaches the $300 goal in three sales. Justify it with a table.	Answers vary; check that 3 × (sales − $50) ≥ $300, i.e., each sale earns at least $150 in sales.

Reflection

#	Prompt	Look for
R1	When can a larger package be a worse buy even with a lower unit rate? Generalize your answer.	When the extra quantity is not needed (or spoils), the total cost is higher; compare total cost for the amount needed, not just unit price.
R2	How do markup and discount percents interact? Is a 20% markup followed by a 20% discount the original price?	No: 1.2 × 0.8 = 0.96, so the price ends 4% lower because the discount is taken from a larger amount.